

FITTRACK AI

Phase 1.2 — Problem Statements

Brainstorming & Ideation Documentation | Source: Main Project Document

1. Document Purpose

This document identifies and structures the core problems addressed by FitTrack AI. The problem statements are derived from the project's documented Background, Problem, Solution, Usage, and Outcome sections. No unsupported business problem or feature has been introduced.

2. Problem Context

The case study describes Rahul as a fitness enthusiast who exercises regularly but records workouts manually using notebooks and spreadsheets. Although he maintains a consistent fitness routine, he experiences difficulty organizing workout history, monitoring progress, and determining whether exercise habits align with fitness goals. The document also identifies a lack of professional guidance when selecting workouts appropriate to his fitness level.

3. Documented Problem Statements

ID	Problem Area	Documented Problem	Documented System Response
PS-01	Centralized workout record problem	Users have difficulty maintaining a centralized record of their workouts.	Provide a centralized backend for creating, retrieving, updating and deleting workout records.
PS-02	Workout history accuracy problem	Manual tracking can lead to inaccurate workout history.	Use structured workout records stored in MongoDB through Mongoose schemas and CRUD APIs.
PS-03	Personal fitness data security problem	The documented problem identifies the absence of secure user authentication for personal fitness records.	Implement registration/login, bcrypt password encryption, JWT authentication and protected APIs.
PS-04	Workout retrieval/search problem	Users are unable to search previous workouts efficiently.	Provide workout search using workout name, category or workout date.
PS-05	Personalized guidance problem	Users lack personalized workout recommendations suited to their fitness information and experience level.	Use Google Gemini AI to generate personalized workout plans, weekly suggestions, recommendations, guidance and safety tips.
PS-06	Performance insight problem	Users lack insights regarding overall workout performance.	Analyze workout statistics and provide AI-generated performance analysis, improvement suggestions, motivational advice and progress summaries.
PS-07	Manual fitness calculation problem	Manual calculation of calories burned and workout duration is time-consuming.	Capture workout duration and calories burned as structured workout information and use workout statistics for AI fitness insights.

4. Consolidated System-Level Problem Statements

ID	Domain	Problem Statement	FitTrack AI Response
CPS-01	Workout Tracking	Fitness users need a reliable centralized mechanism to record and manage workout history instead of depending on manual notebooks or spreadsheets.	Workout CRUD APIs with authenticated user ownership and MongoDB persistence.
CPS-02	Secure Access	Personal fitness records need controlled access so that workout information is available through authenticated APIs.	JWT-based authentication, bcrypt password hashing and authentication middleware.
CPS-03	Workout Discovery	Users need an efficient way to locate historical workout records using meaningful workout attributes.	Search by workout name, category or workout date.
CPS-04	Personalized Exercise Guidance	Users need fitness guidance that considers age, fitness goal and workout experience.	Google Gemini integration for personalized workout recommendations and weekly exercise suggestions.
CPS-05	Fitness Performance Understanding	Users need assistance interpreting workout statistics and understanding potential areas for improvement.	AI fitness insights based on total workouts, average duration and total calories burned.

5. Problem-to-Feature Traceability

Problem	Relevant Feature / Module	Expected Contribution
Manual / decentralized workout records	Workout Management + MongoDB	Structured persistent workout history.
Inaccurate manual history	Mongoose schemas + validation	Structured fields and documented validation rules.
Lack of secure authentication	Authentication + JWT Middleware + bcrypt	Authenticated access to protected APIs.
Inefficient workout search	Workout Search	Search by name, category or date.
No personalized recommendations	Gemini AI Recommendation	Personalized plans, exercise suggestions, training guidance and safety recommendations.
No performance insights	AI Fitness Insights	Performance analysis, improvement suggestions, motivational advice and progress summary.
Time-consuming manual calculations	Workout data + AI Fitness Insights	Store duration/calories and analyze supplied workout statistics.

6. Scope Boundary

The main document identifies nutrition tracking, BMI calculation, wearable-device integration, workout reminders and fitness analytics dashboards as possible future extensions. These are not treated as current problems or current implemented solutions in this document.

Prepared exclusively from the supplied Main Project Document. Problem statements are structured from documented evidence and do not introduce unsupported functionality.